# Audit Trail — Case 11: Shoulder closure on divided highway

| Field | Value |
|---|---|
| Plan status | 2 changes · 1 needs attention · 12 checked · 1 pending · reference |
| MUTCD typical application | TA-3 |
| CDOT standard sheet | S-630-1 |
| Case | Case 11: Shoulder closure on divided highway |
| Taper length | 150 ft (L/3 (shoulder taper)) |
| Buffer space | 360 ft |
| Device spacing — taper | 45 ft |
| Device spacing — tangent | 90 ft |
| Crew steps | 14 |
| Routing | shoulder_no_reduction |

## Taper Length

Speed 45 mph >= 45 mph threshold -> using L = W x S

L = 10 x 45 = 450 ft

L/3 = 450 / 3 = 150 ft

| Field | Value |
|---|---|
| Closure type | shoulder |
| Offset (shoulder width) | 10 ft |
| Required taper (L/3 (shoulder taper)) | 150 ft |

Source: MUTCD 11th Ed. Sec 6B.08, Table 6B-4 (taper length L). Shoulder closures use L/3 per Sec 6B.08 (Table 6B-3).

CDOT reference: CDOT S-630-1 Case 11 (right-shoulder closure on divided highway)

## Buffer Space

MUTCD Table 6B-2: 360 ft (CDOT supplement silent at this speed)

Source: MUTCD 11th Ed. Sec 6B.06, Table 6B-2 (stopping sight distance)

## Channelizing Device Spacing

45 mph -> 45 ft spacing (MUTCD Sec 6K.01: spacing equals speed in feet)

45 mph -> 90 ft spacing (MUTCD Sec 6K.01: spacing equals 2x speed in feet)

L/3 = 150 ft / 45 ft max spacing -> 5 drums at 37.5 ft intervals

1000 ft / 90 ft max spacing -> 13 cones at 83.3 ft intervals

Downstream taper: 50 ft at ~25 ft spacing -> 2 cones (MUTCD Sec 6B.08: 50-100 ft downstream taper); 13 tangent + 2 downstream = 15 cones total

Source: MUTCD 11th Ed. Sec 6K.01

## Advance Warning Signs

Speed 45 mph, road_type = 'urban_high' -> Table 6B-1 category: urban_high

A = 350 ft, B = 350 ft, C = 350 ft

| Position | Code | Station (ft) | Distance from Taper (ft) |
|---|---|---|---|
| C (furthest) | W20-1 | 2,560 | 1,050 upstream |
| B (middle) | W20-2 | 2,210 | 700 upstream |
| A (nearest) | W21-5aR | 1,860 | 350 upstream |

Source: MUTCD 11th Ed. Table 6B-1

## Colorado Requirements (CDOT S-630-1)

| Check | Result | Citation | Detail |
|---|---|---|---|
| Signs on both sides of divided highway | FAIL | CDOT S-630-1 (July 2026) Sheet 2, General Note 8 | Required: True. Signs placed: 6 left, 10 right. |
| G20-5P Work Zone signs every 2,640 ft | PASS | CDOT S-630-1 (July 2026) Sheet 2, General Note 4 | Zone length: 2,560 ft. Required: 1. Placed: 1. |
| Speed reduction <= 15 mph per sign installation | PASS | CDOT S-630-1 (July 2026) Sheet 2, General Note 3 | No work-zone speed reduction. Posted speed 45 mph applies throughout the zone. |
| Flagger station lighting 500W @ 8 ft | PASS | CDOT S-630-1 (July 2026) Sheet 2, General Note 22 | Not applicable (no flaggers). |

AADT threshold for mobile operations (<= 2,000) — Not applicable (not a mobile operation). (CDOT S-630-1 (July 2026) Sheet 23, Case 38 Note 1)

All Colorado checks pass: False

## CDOT Case Reference

Case 11: Shoulder closure on divided highway

This scenario matches CDOT Standard Plan S-630-1, Case 11: shoulder closure on a divided highway.

The generated plan follows the same device layout as the reference case with spacing computed for 45 mph.

Routing: shoulder_no_reduction

Reference:
https://www.codot.gov/safety/traffic-safety/assets/s-standard-plans/s-630-1/S-630-01%20(19-Page%20Set).pdf

## Geometry Validation

Work zone 1000 ft vs required taper 150 ft (L/3 (shoulder taper)) + buffer 360 ft.

All geometry checks pass: True

Source: MUTCD 11th Ed. Sec 6B.06 (buffer) and Sec 6B.08 (taper)

## Corridor Validation

Corridor check unavailable — OpenStreetMap could not be reached at generation; road-network warnings were not evaluated. Re-generate to retry.

## Site Conditions

Scanned along the corridor against OpenStreetMap at 2026-09-04T00:00:50+00:00 (16419 ms, memoised). The five rule-bearing conditions below are plan facts; a detected condition fired the matching Site Adjustment that follows.

| Condition | Result | Evidence |
|---|---|---|
| Adjacent at-grade intersection | DETECTED | 26 found · nearest 34.1 ft from anchor · unnamed at 39.7038, -105.0816 [outside @ 2732 ft] |
| Adjacent interchange (highway ramps) | None along the corridor | |
| Pedestrian sidewalks | DETECTED | 18 found · nearest 46.6 ft from anchor · unnamed at 39.7112, -105.0812 [lateral 75 ft off centerline] |
| Bike lane / cycleway | DETECTED | 27 found · nearest 71.2 ft from anchor · unnamed at 39.7092, -105.0818 [lateral 78 ft off centerline] |
| School zone | None along the corridor | |
| Railroad crossing | Reference — none | |
| Hospital | Reference — none | |
| Road curvature | Reference — none | |

## Site Adjustments

Site-condition flags layered onto the baseline MUTCD/CDOT layout. Each adjustment is traced to its source rule; the rendered plan, device list, and crew narrative reflect every item below.

| Rule | Adjustment |
|---|---|
| MUTCD §6N.12 p. 848 — Work within the Traveled Way at an Intersection (11th Ed.) | No devices added — the cross-street approach layout is not generated; see the pending-verification disclosure. |
| MUTCD §6C.02 — pedestrian considerations in work zones | Added 4 Type III barricades (sidewalk closure points) and 2 R9-9 SIDEWALK CLOSED signs at the upstream and downstream ends of the work zone. |
| MUTCD §9C.101 — bicycle facility work zone signing | Added 2 M4-9a BIKE DETOUR signs at the upstream and downstream ends. |

## Flagger Control

Not applicable for this scenario.

Source: MUTCD 11th Ed. Chapter 6D (Flagger Control) and Chapter 6E (One-Lane, Two-Way Traffic Control).

## Pending Verification

1 reference(s) pending verification. An adjacent intersection is flagged, but Conestruct does not generate the cross-street approach layout or evaluate traffic-signal operation, and adds no devices for it. Cross-street control design and the signal-operation review (MUTCD §6N.12, 11th Ed. p. 848; Part 4 for signal-head visibility) remain with the traffic control supervisor. Intersection support is tracked at issue #117; corner-quadrant work at issue #128. Tracking: https://github.com/rtmakatura/conestruct/issues/117

- [intersection_layout_not_generated] An adjacent intersection is flagged, but Conestruct does not generate the cross-street approach layout or evaluate traffic-signal operation, and adds no devices for it. Cross-street control design and the signal-operation review (MUTCD §6N.12, 11th Ed. p. 848; Part 4 for signal-head visibility) remain with the traffic control supervisor. Intersection support is tracked at issue #117; corner-quadrant work at issue #128. (https://github.com/rtmakatura/conestruct/issues/117)